SHREYASH V. THORAT

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EDUCATION

University of Southern California Los Angeles, CA.
Master of Science in Mechanical Engineering August 2025-May 2027

- Hardware Engineer: **The Laboratory for Autonomous Systems in Exploration and Robotics (**LASER) with Professor Keenan Albee, working on a Microgravity Robot designed to be operated in uncertain environment.

K. J. Somaiya College of Engineering Mumbai, India
Bachelors of Technology in Mechanical Engineering July 2020-January 2024

- Minors' degree in Artificial Intelligence & Machine Learning
- Coursework: Robotics and Automation, Mechatronics, Computer Aided Engineering, Additive Manufacturing

RELEVANT SKILLS

- Software: Python, C++, Linux
- Robotics: ROS2, Gazebo, Linux, Git, RTOS, PID & MPC Control, MATLAB/SIMULINK, Motion Planning (Particle Filter, RRT)
- Mechanical: SolidWorks, ANSYS (ACP, Workbench), Catia V5,

EXPERIENCE

Space Engineering Research Center: Information Sciences Institute (USC) Los Angeles, CA
Graduate Student Researcher Sept 2025-Present

- Collaborating with hardware and propulsion team on the Leapfrog Lunar Lander prototype, analyzing test results, and proposing design iterations to improve landing stability

Greendzine Technologies Pvt Ltd Bangalore, India
Mechanical Design Engineer January 2024-May 2024

- Engineered and prototyped mechanical components for warehouse intra-logistics electric vehicles (100,000+ sq. ft. facilities), applying DFMA principles to improve manufacturability and reduce assembly time

Orion Racing India | Formula Student Team Mumbai, India
Suspension & Steering Subsystem Lead Feb 2021-Dec 2023

- Led a 12-member subsystem team through design reviews, sprint planning and working with departments to validate integration points, enhance design and resolve cross-system mechanical issues
- Analyzed data acquisition logs to correlate real-world track performance with theoretical vehicle model to tune the system to boost driver performance.
- Awards: 2nd Position at Formula Bharat Electric Category, 2nd in Cost & Manufacturing at FSA-Austria

ACADEMIC PROJECTS

5-DOF Package Handling Robotic Arm for Pick and Place Operations Mumbai, India
Independent Project September 2024-January 2025

- Designed a high-fidelity 5-DOF manipulator (700mm reach, 3kg payload) in SolidWorks, performing Inverse Kinematics (IK) analysis to define workspace limits and singularity avoidance.
- Integrated a ROS 2 control stack with MoveIt, utilizing motion planning to execute efficient pick & place trajectories in Gazebo.

Model Rocket Development with Active Thrust Vector Control Mumbai, India
Project Lead July 2024-December 2024

- Implemented active thrust vectoring using servo motors and a gyroscope sensor for flight stability; integrated a servo-actuated parachute system for recovery
- Conducted ground and static tests to validate thrust vectoring response, recovery deployment reliability, and overall system stability under launch conditions achieving 384 N peak thrust and 320 N-s total impulse

Real Time Surveillance Quadcopter for Flood Rescue Operation Mumbai, India
Project Lead June 2023-September 2023

- Performed MATLAB flight simulations to analyze power consumption, ESC efficiency, and battery performance, enhancing system endurance and reliability
- Implemented a MPC algorithm within a physics-based MATLAB simulation, utilizing a 6-DOF dynamic model to minimize state error and control effort.